AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

Computer Science Paper II

Time: 2 hours 20 minutes Marks: 55

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. After Question 4, there are TWO sections, section I on C Language and section II on Visual Basic. Attempt EITHER section.
   In each section answer ALL questions. The last question in each section offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. A database object is used to store or reference data.

Name any THREE database objects. (3 Marks)

b. A database table consists of fields and records.

Describe the major difference between a field and a record. (2 Marks)

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Q.2. (Total 5 Marks)

Omar has created a database table that is shown below.

Student ID	First Name	Last Name	Class	Date of Birth	Fees Paid
S101	Ali	Khan	11	10/09/1999	Yes
S102	Jahangir	Ahmed	11	03/15/1998	No
S103	Sumaira	Usmani	12	06/26/1997	No
S104	Nadia	Zafar	12	03/15/1998	Yes
S105	Ali	Ahmed	12	11/20/1997	Yes

a. Write the CORRECT data types he selected for the fields given below: (3 Marks)

Student ID: _____

Class: _____

Fees Paid: _____

b. Write the name of the field from the given table which should be selected as a primary key. Give a valid reason to support your answer. (2 Marks)

Primary Key: _____

Reason: _____

_____

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Q.3. (Total 4 Marks)

a. Differentiate between the two database options filter and sorting. (2 Marks)

b. Consider the given table.

Microsoft Access - [Employees : Table]

File Edit View Insert Format Records Tools Window Help

Employee ID	Employee Name	Salary	City	Birthdate
1	Kamal Ahmed	$25,000.00	Karachi	12/2/1978
2	Asad Shah	$40,000.00	Islamabad	3/5/1976
3	Ali Khan	$30,000.00	Lahore	4/6/1980
4	Syed Faisal	$34,000.00	Karachi	1/1/1979
5	Amjad Khan	$50,000.00	Karachi	12/2/1978
6	Kamran Khan	$70,000.00	Lahore	1/27/1977
(AutoNumber)				

i. If the table is sorted in ascending order by "Employee Name", then which employee's record will be mentioned at the end? (1 Mark)

ii. If the table is sorted in descending order by "Birthdate", then which employee's record will be mentioned in the beginning? (1 Mark)

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Q.4. (Total 6 Marks)

A database named 'CarpetRecords' was setup to store data of carpets such as type, colour, material, quantity, price, etc. Some records from the database are shown below.

CarpetRecords						
Batch ID	**Type of Carpet**	**Quantity**	**Colour**	**Material Used**	**Made in**	**Price in Rupees**
C100	Woven	5	Black	Wool	Pakistan	18000
C101	Needle Felt	3	Brown	Polyester	China	12500
C102	Needle Felt	5	Blue	Wool	Pakistan	20000
C103	Tufted	7	Green	Nylon	Turkey	13000
C104	Woven	6	Blue	Wool	China	17000
C105	Needle Felt	2	Black	Wool	Turkey	22500

i. Consider the following query in design view of MS Access. (3 Marks)

Field:	Batch ID	Type of Carpet	Colour	Made in	Price in Rupees
Table:					
Sort:					
Show:	☐	☑	☑	☐	☑
Criteria:			= "Black"		
or:				= "Pakistan"	

Show the output when the given query is executed.

Space for output

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ii. Complete the query in design view shown below to select and show the Batch ID, Type of Carpet, Quantity, Made in and Price in Rupees of all types of carpets with 5 or more quantity.
(3 Marks)

Field:					
Table:					
Sort:					
Show:	☐	☐	☐	☐	☐
Criteria:					
or:					

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SECTION I

C Language

Q.5. (Total 5 Marks)

a. Define the term 'high level programing language'. (1 Mark)

b. Describe FOUR advantages of C programing language over other programing languages. (4 Marks)

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Q.6. (Total 5 Marks)

A student needs a computer program to convert temperature from degrees Fahrenheit (°F) to degrees Celsius (°C). Write a program using C language for the student that would meet the following requirements:

- It should take degrees Fahrenheit value as input.
- The degrees Fahrenheit value should be converted into degrees Celsius using the formula $°C = (F - 32) \div 1.8$
- It should display degrees Celsius value with appropriate message in output.

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Q.7. (Total 5 Marks)

Write a program using C language which will take two integer values as input, test these two values to check if the first value is evenly divisible (no remainder left) by the second value and, finally, display appropriate output message.

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Q.8. (Total 6 Marks)

a. State the purpose of the following elements of functions in programming.

i. Function prototype (1 Mark)

ii. Function call (1 Mark)

iii. Function definition (1 Mark)

b. Write the TWO purposes of return statement in C functions? (2 Marks)

c. In a computer language, there are two ways in which arguments can be passed to a function, call by value and call by reference.

Which of these is the default way in C language for passing arguments to a function? (1 Mark)

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Q.9. (Total 4 Marks)

a. Differentiate between getc() and putc() functions. (2 Marks)

b. Write the C language statement to open a text file named mytextfile.txt in write mode. (2 Marks)

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Q.10. (Total 10 Marks)

EITHER

a. Write a program in C language that would meet the following criteria.

- It should input marks for three subjects, i.e. Physics, Chemistry and Mathematics.
- The marks should be between 0 and 100 (both inclusive) and if any marks value entered is outside of this range, then an error message should be displayed.
- It should calculate and display the average marks of the three subjects.
- It should display relevant remarks depending upon the average marks of the three subjects. (See table)

Average Marks	Remarks
80 to 100	Excellent
60 to 79	Good
50 to 59	Satisfactory
0 to 49	Needs Improvement

OR

b. A gym trainer wants to calculate the average weight of the gym members.

Write a program using C language that would input weight of 35 gym members. The program should also calculate and display the maximum weight and average weight values of all gym members.

(**Note:** Use the For loop for repeating the code.)

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SECTION II

Visual Basic

Q.5. (Total 5 Marks)

a. With reference to Visual Basic, differentiate between objects and methods and support your answer with an example for each. (4 Marks)

b. Define the term ‘event driven programing’. (1 Mark)

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Q.6. (Total 5 Marks)

Omar has made a calculator using a Form in Visual Basic which contains:

- three text boxes named as **tbox1**, **tbox2**, **tbox3**
- four buttons with captions **Add**, **Subtract**, **Multiply** and **Divide**.

The calculator will input two numbers, i.e. one in **tbox1** and the second in **tbox2** and then one of the four buttons will be clicked to perform the corresponding arithmetic operation and the result will be shown in **tbox3**.

Write the Visual Basic code for the Division button to perform the operation of division and declare the required variable.

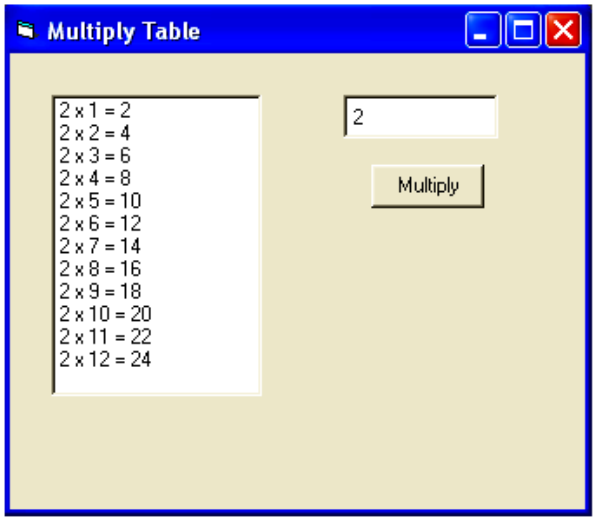
- Declaration of Variables

- Code for Division Button

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Q.7. (Total 5 Marks)

Consider the given Form made in Visual Basic.



This Form will get an input number and generate its table to a maximum of 12 times. Write the code for Multiply button to generate the table and display it in the list of form.

(**Note:** Use Do While Loop to solve the given problem. The name of text box is **TB1** and text list is **TL1**.)

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Q.8. (Total 6 Marks)

a. Describe the purpose of master view in database programing and give an example. (3 Marks)

b. Differentiate among snapshot, dynaset and recordset. (3 Marks)

Snapshot	Dynaset	Recordset

Q.9. (Total 4 Marks)

a. State any TWO purposes of using aspect ratio in the Circle method of Visual Basic. (2 Marks)

b. List the TWO benefits of using ActiveX Data Object (ADO) over data control. (2 Marks)

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Q.10. (Total 10 Marks)

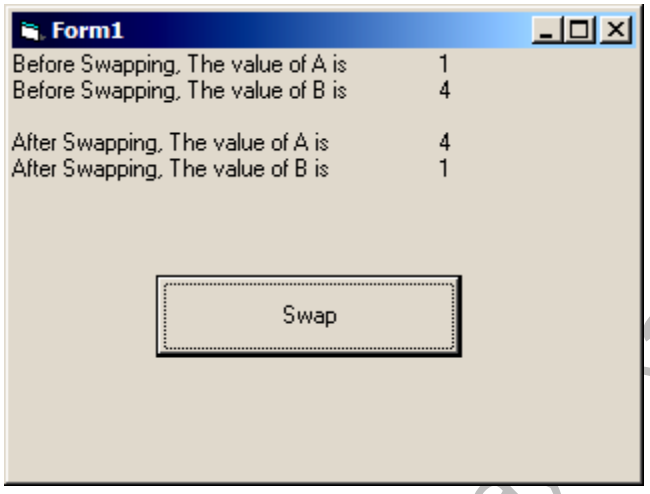
EITHER

a. Write a Visual Basic program that would generate 100 random numeric values between the range of 1 and 20 in an array and print these values in the ascending order.

OR

b. Write a Visual Basic program that would input two values and store them in variable **X** and variable **Y**. The program should also swap the values of **X** and **Y** and display variable values before and after swapping.

(**Note:** The example is given for your reference.)



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END OF PAPER

Please use this page for rough work

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